

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	0.55	Gender	Male
Height (cm)	172.0	Weight (kg)	88.0
BMI	29.7	Product Info 2	D2
Occupation	Sales Director	Annual Income	\$1,500,000.00
Insurance Type	Life	Coverage Amount	\$9,000,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 1 of 8
Risk Category	Low Risk
Recommended Annual Premium	\$10,000.00

3. AI Actuarial Analysis

Actuarial Report: Insurance Application Evaluation

****Applicant Details:****

- * Age: 0.55 (Critical Anomaly)
- * Height: 172.0 cm, Weight: 88.0 kg, BMI: 29.7 kg/m²
- * Occupation: Sales Director
- * Income: \$1,500,000.0
- * Smoker: Yes
- * Previous Claims: 2
- * Family History: Positive
- * Insurance Type: Life, Coverage Amount: \$9,000,000.0
- * Lifestyle: Exercises, Moderate Alcohol
- * Gender: Male

****Machine Learning Classifier Output:****

- * Predicted Risk Class: 1 (Lower is better)
- * Prediction Probability/Confidence: 68.0%
- * Calculated Annual Premium: \$10,000.00

1. Risk Assessment

The applicant presents a complex risk profile with several high-impact factors:

* ****Age:**** The reported age of 0.55 years is a critical data anomaly. This indicates either an infant, which is inconsistent with all other demographic (height, weight, BMI), occupational, and income data, or a significant data entry error for an adult applicant. This requires immediate clarification and correction. Assuming, for the purpose of assessing other factors, that this is an error and the applicant is an adult, we proceed to other factors.

* ****BMI (29.7):**** This falls into the "overweight" category, bordering on "obese" (BMI $\geq$ 30). This is a moderate health risk factor for various conditions.

* ****Smoking Status (Smoker=1):**** This is a significant adverse risk factor, leading to higher mortality and morbidity rates across all ages.

* ****Previous Claims (2):**** A history of two previous claims is a substantial negative risk indicator. It suggests a higher propensity for future claims and requires detailed review of the nature and recency of these claims.

* ****Family History (Positive=1):**** A positive family history indicates a genetic predisposition to certain medical conditions, increasing the applicant's risk.

* ****Gender (Male):**** Statistically, males generally have higher mortality rates than females for life insurance purposes.

* ****Occupation (Sales Director):**** Generally a professional role, which often implies lower physical risk. However, it can be high-stress, potentially impacting health long-term.

* ****Income (\$1,500,000.0) & Coverage Amount (\$9,000,000.0):**** The coverage amount is 6 times the reported income. While within plausible limits for high-income individuals, it is at the higher end and, combined with other adverse risk factors, warrants scrutiny to ensure insurable interest and financial justification.

* ****Lifestyle (Exercise=1, Alcohol=2):**** Exercise is a positive mitigating factor. Moderate alcohol consumption is

generally neutral to slightly positive.

Overall, considering the prominent risk factors (smoker, previous claims, family history, high BMI, male gender), the applicant appears to present an elevated risk profile.

2. Comparative Analysis

The current application's predicted Risk Class 1 and premium of \$10,000.00 can be compared with the similar historical cases:

* **Application POL-8902 (Similarity: 99.8%, Risk Class: 2, Premium: \$12,000.00, Approved):** This case is extremely similar to the current applicant but was assigned a slightly higher Risk Class (2) and a higher premium (\$12,000.00). This suggests the current applicant, despite high similarity, is either marginally better on specific unstated attributes or the model has identified a slightly different risk profile that allows for a lower risk class.

* **Application POL-4519 (Similarity: 98.3%, Risk Class: 4, Premium: \$17,000.00, Approved):** This case, while highly similar, was assigned a significantly higher Risk Class (4) and premium (\$17,000.00). The current application differs significantly here, being rated as much lower risk. This could be due to specific differentiating factors not visible in the provided data (e.g., severity of previous claims, precise BMI difference, smoking duration).

* **Application APP-0642B5 (Similarity: 86.3%, Risk Class: 1, Premium: \$10,000.00, Approved):** This case is an exact match in terms of Risk Class and Premium. Although its similarity score is lower than the other two, it directly supports the ML model's output for the current application. This suggests the model has found a strong precedent for this specific risk-premium combination.

The ML model's output for the current applicant (Risk Class 1) is notably more favorable than POL-8902 (Risk Class 2) despite higher similarity to POL-8902, and matches APP-0642B5. This suggests the model has identified specific (potentially unstated) features that lead to the lowest risk classification for the current applicant, but this needs manual verification given the observed high-risk factors.

3. Anomaly & Fraud Detection

* **Critical Age Anomaly:** The most significant anomaly is the applicant's age (0.55 years). This is inconsistent with all other provided data (height, weight, BMI, occupation, income) and must be a data entry error. This discrepancy requires immediate investigation and correction, as the entire risk assessment hinges on the correct age of the applicant.

* **Discrepancy in Risk Class vs. Factors:** There is a significant apparent contradiction between the predicted Risk Class 1 (lowest risk) and the applicant's multiple high-risk factors (smoker, 2 previous claims, positive family history, high BMI, male gender). This low risk classification, with only 68% confidence, is an anomaly that warrants deep manual review. It raises questions about the model's feature interpretation or potential data quality issues in the training set if similar high-risk profiles were incorrectly classified as low risk.

* **High Number of Previous Claims:** Two previous claims are a notable red flag. The nature, severity, and timing of these claims need to be fully understood, as they significantly influence future claim probability and policy eligibility.

* **Coverage Amount vs. Income:** While a \$9,000,000.0 coverage amount for a \$1,500,000.0 income (6x) is not inherently fraudulent, it is on the higher side. Combined with the other high-risk factors and the questionable low-risk classification, it necessitates verification of insurable interest and financial justification to prevent anti-selection.

4. Premium Recommendation

The calculated annual premium of \$10,000.00 aligns perfectly with the predicted Risk Class 1 and the highly similar historical case APP-0642B5 (Risk Class 1, Premium \$10,000.00, Similarity 86.3%).

However, the justification for this premium is *severely undermined* by the anomalies identified:

1. **Critical Age Anomaly:** Until the correct age is established, any premium calculation is hypothetical.
2. **Discrepancy in Risk Class:** The predicted Risk Class 1 appears highly optimistic given the prominent adverse risk factors (smoker, 2 previous claims, positive family history, high BMI, male gender). The model's 68% confidence level is not exceptionally strong for such a favorable prediction under these circumstances.

Therefore, while the premium aligns with the model's output, its robustness is highly questionable until the underlying risk assessment (Risk Class 1) is validated manually. A more conservative premium might be expected given the presented risk profile.

5. Underwriting Decision Recommendation

Recommendation: Refer for Manual Underwriting Review with High Priority.

This application cannot be approved or rejected based solely on the current data and ML output due to several critical issues:

1. **Critical Data Error (Age):** The reported age of 0.55 years is a fundamental data anomaly. The application cannot proceed until the applicant's correct age is verified.
2. **Inconsistent Risk Assessment:** The ML model's prediction of Risk Class 1 (lowest risk) contradicts the significant adverse risk factors present (smoker, 2 previous claims, positive family history, high BMI, male gender, high coverage amount). This discrepancy, coupled with a moderate confidence score (68%), necessitates a thorough manual review to validate the actual risk level.
3. **High Previous Claims:** Two previous claims require detailed investigation into their nature, severity, and impact on future mortality/morbidity.
4. **High Coverage Amount:** The \$9,000,000.0 coverage requires careful financial underwriting to confirm legitimate insurable interest and affordability given the income and overall risk profile.

Specific Actions for Manual Review:

- * Verify and correct the applicant's age.
- * Obtain detailed information regarding the two previous claims (dates, causes, payouts).
- * Request attending physician statements (APS) for a comprehensive health assessment, particularly regarding the high BMI, smoking history, and family history.
- * Conduct a financial needs analysis to justify the \$9,000,000.0 coverage amount.
- * Re-evaluate the overall risk profile and determine an appropriate risk class and premium based on verified and comprehensive data. It is highly probable that the applicant's true risk class is higher than the ML model's prediction of 1, and the corresponding premium may need adjustment.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$10,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.